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# Sheaf cohomology

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# Introduction

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# Chapter 1

# Presheaves and sheaves

## 1.1 Presheaves

We begin by giving a general definition of a presheaf.

**Definition 1.1.1** (Presheaf)**.** Let $X$ be a topological space and let $\mathsf{Op}(X)$ be its Posetal category of open sets. Let $\mathsf{C}$ be a `def:small-category`. The Category of $\mathsf{C}$-valued **presheaves** is the `ex:functor-category`

$$\mathsf{PSh}_X(\mathsf{C}) := [\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C}]$$

When the topological space is clear we will write $\mathsf{PSh}(\mathsf{C})$.

The elements of this category are Functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are Natural transformations $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Let's rewrite Definition 1.1.1 less abstractly: a $\mathsf{C}$-valued Presheaf $\mathcal{F}$ is the collection of data

- For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ of $\mathsf{C}$ called *sections of $\mathcal{F}$ over $U$*.
- For every inclussion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in $\mathsf{C}$ called *restriction* such that
  - For every open set $U$, $\mathrm{res}_{U,U} = \mathrm{id}_U$.
  - For every inclussion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A *morphism of Presheaves* $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ in $\mathsf{C}$ making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

We are going to study now Limits and Colimits in $\mathsf{PSh}(\mathsf{C})$.

Note that, since $\mathsf{PSh}(\mathsf{C})$ is a `def:functor-category` then if $\mathsf{C}$ has Limits or Colimits of a certain shape then $\mathsf{PSh}(\mathsf{C})$ also has them and they are computed pointwise.

This means that Limits and Colimits in $\mathsf{PSh}(\mathsf{C})$ can be computed *open set by open set*.

**Proposition 1.1.1.** *Let $X$ be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a Diagram. If $\mathsf{C}$ has Limits and/or Colimits with shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ and/or $\operatorname{colim}_i \mathcal{F}_i$ exist in $\mathsf{PSh}_X(\mathsf{C})$. Moreover for every open set $U \subseteq X$*

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \mathcal{F}_i(U) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \mathcal{F}_i(U)$$

Write a section on functor categories (or quote it from somewhere) to justify this.

In particular if $\mathsf{C}$ is an Abelian category then Kernels and Cokernels exist and are computed *open set by open set*, making $\mathsf{PSh}(\mathsf{C})$ an Abelian category.

*Remark* 1.1.1. From this point onwards, we will suppose $\mathsf{C}$ has all Limits and Colimits unless stated otherwise. ▽

### 1.1.1 Presheaves and continuous maps

**Definition 1.1.2** (Direct image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a $\mathsf{C}$-Presheaf over $X$. The **direct image Presheaf** of $\mathcal{F}$ via $f$ is the Presheaf $f_*\mathcal{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(U))$$

Given $W \subseteq V \subseteq Y$ open subsets, the restriction map comes from the commutativity of the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow \\ f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \end{array}$$

This construction defines a Functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Show that the category of presheaves over a one point space *is equal* to the evaluation category.

**Example 1.1.1** (Global section)**.** Let $X$ be a topological space. Consider the one-point topological space $\{*\}$. Let $\mathcal{F}$ be a $\mathsf{C}$-valued presheaf over $X$. The **global section** of $\mathcal{F}$ over

This defines a Functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$ ◊

**Example 1.1.2** (Skyscrapper Presheaf)**.** Let $X$ be a topological space and $p \in X$. Suppose $\mathsf{C}$ has a `def:final-object` $* \in \mathsf{C}$. Let $S$ be an object in $\mathsf{C}$. The **skyscrapper Presheaf** supported at $S$ $\iota_{p,*}\underline{S}_{\text{pre}}$ is the Direct image Presheaf of the Constant Presheaf $\underline{S}_{\text{pre}}$ via the inclussion $\iota : \{p\} \hookrightarrow X$:

$$(\iota_{p,*}\underline{S}_{\text{pre}})(U) = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

◊

Using this we can construct a *dual* Presheaf to Definition 1.1.2.

**Definition 1.1.3** (Inverse image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{G}$ be a $\mathsf{C}$-valued Presheaf over $Y$. The **inverse image Presheaf** of $\mathcal{G}$ via $f$ is the Presheaf $f^{-1}\mathcal{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathcal{G}(U) = \operatorname*{colim}_{V \supseteq f(U)} \mathcal{G}(V)$$

where the Colimit is taken over all the open neighbourhoods $V$ of $f(U)$.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

*Remark* 1.1.2. Note that the set of open neighbourhoods of $f(U)$ is a partially ordered set via reverse inclussion. Therefore if $\mathsf{C}$ is a concrete category we can describe $f^{-1}\mathcal{G}(U)$ as $\{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y), s \in \mathcal{G}(V)\}$ modulo the relation $(s, V) \sim (s', V')$ if and only if there exists $W \subseteq V \cap V'$ open with $f(U) \subseteq W$ such that $s|_W = s'|_W$. ▽

**Example 1.1.3** (Constant Presheaf)**.** Let $X$ be a topological space. Let $S$ be an object of $\mathsf{C}$. The **constant Presheaf** associated to $S$ $\underline{S}_{\text{pre}}$ is the Inverse image Presheaf of $S$ via the map $\{*\} \to X$. ◊

**Definition 1.1.4** (Stalk)**.** Let $X$ be a topological space and $p \in X$. Let $\mathcal{F}$ be a $\mathsf{C}$-valued Presheaf over $X$. The **stalk** of $\mathcal{F}$ at $p$ $\mathcal{F}_p$ is the Inverse image Presheaf of $\mathcal{F}$ via $X \to \{p\}$:

$$\mathcal{F}_p = \operatorname*{colim}_{U \ni p} \mathcal{F}(U)$$

⌟

**Proposition 1.1.2.** *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Presheaf functor and the Inverse image Presheaf $f^{-1} \dashv f_*$.*

*Proof.* We define the `def:categorical-unit` $\eta : 1 \implies f_* f^{-1}$ as the map with components given by the canonical insertion maps to the colimit

$$\mathcal{F}(W) \to \operatorname*{colim}_{U \supseteq (f \circ f^{-1})(W)} \mathcal{F}(U) = f^{-1}\mathcal{F}(f^{-1}W) = f_* f^{-1}\mathcal{F}(W)$$

Note that this maps exist because $(f \circ f^{-1})(W) \subseteq W$.

On the other hand, note that if $f(V) \subseteq U$ then $V \subseteq (f^{-1} \circ f)(V) \subseteq f^{-1}(U)$, i.e: we have a restricion map $\mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$. Therefore by the universal property of the colimit we have a unique map

$$f^{-1} f_* \mathcal{G}(V) = \operatorname*{colim}_{U \supseteq f(V)} \mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$$

Using this maps we obtain a `def:categorical-counit` $\varepsilon : f^{-1} f_* \implies 1$.

Since by construction the compositions $\eta \circ \varepsilon$ and $\varepsilon \circ \eta$ are the corresponding identity morphisms, we have proven the adjunction. □

Two important particular cases of the adjunction Proposition 1.1.2 are

**Corollary 1.1.1.** *Let $X$ be a topological space and $p \in X$. Let $\mathcal{F}$ be a $\mathsf{C}$-valued Presheaves over $X$ and let $A$ be an object in $\mathsf{C}$. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{A}_{\text{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathcal{F}))$$

**Corollary 1.1.2.** *Let $X$ be a topological space and $p \in X$. Let $\mathcal{F}$ be a $\mathsf{C}$-valued Presheaves over $X$ and let $A$ be an object in $\mathsf{C}$. Then*

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, A) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{A}_{\text{pre}})$$

## 1.2 Sheaves

We now turn our attention to a particular class of Presheaves, following [**stacks-project**].

Let's start by considering the case $\mathsf{C} = \mathsf{Set}$:

**Definition 1.2.1** (Sheaf of sets)**.** Let $X$ be a topological space. A Presheaf of sets $\mathcal{F}$ is a **sheaf** if for every open set $U$ and open covering $\{U_i\}_{i\in\mathcal{I}}$ of $U$ the diagram

$$\mathcal{F}(U) \to \prod_{i\in\mathcal{I}} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathcal{F}(U_i \cap U_j) \tag{1.2.1}$$

is an Equalizer.

We denote by $\mathsf{Sh}_X(\mathsf{Set})$ (or just $\mathsf{Sh}$ if the context is clear) the Full Subcategory of the category of presheaves of sets whose objects are sheaves of sets.

Less abstractly this means that $\mathcal{F}$ verifies:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}\ \ s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}\ \ s|_{U_i} = s_i$.

⌟

**Definition 1.2.2** (Category of Sheaves of sets)**.** The Category $\mathsf{Sh}_X(\mathsf{Set})$ of Sheaves of sets is the Full Subcategory of the category of Presheaves $\mathsf{PSh}_X(\mathsf{Set})$ whose objects are Sheaves of sets. ⌟

Let's try to make Definition 1.2.1 more general by considering another evaluation category $\mathsf{C}$.

Firstly, note that if Products indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ for any indexing set $\mathcal{I}$ exist in $\mathsf{C}$ then the diagram (1.2.1) makes sense. Secondly, we have that (1.2.1) is an Equalizer if and only if for every object $C \in \mathsf{C}$ the diagram in $\mathsf{Set}$

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U)) \to \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i \cap U_j))$$

is an Equalizer.

Therefore we get

**Definition 1.2.3** (Sheaf)**.** Let $X$ be a topological space. Let $\mathsf{C}$ be an arbitrary category. A $\mathsf{C}$-valued Presheaf $\mathcal{F}$ is a **sheaf** if for all objects $C \in \mathsf{C}$ the Presheaf of sets $\mathcal{F}_C$ defined as $\mathcal{F}_C(U) = \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U))$ is a Sheaf of sets.

We denote by $\mathsf{Sh}_X(\mathsf{C})$ the Full Subcategory of the category of $\mathsf{C}$-valued presheaves of sets whose objects are $\mathsf{C}$-valued sheaves. ⌟

*Notation.* From now on, to distinguish a Presheaf from a Sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

By Definition 1.2.3 we have an inclussion functor $\iota : \mathsf{Sh}(\mathsf{C}) \hookrightarrow \mathsf{PSh}(\mathsf{C})$.

For *sufficiently nice* Categorys, this Functor admits a left-adjoint $\mathsf{PSh}(\mathsf{C}) \to \mathsf{Sh}(\mathsf{C})$. Let's study first the case $\mathsf{C} = \mathsf{Set}$.

How general do we want to go?

**Theorem 1.2.1** (Sheafification). *There exists a* ***sheafification*** *Functor* $(-)^{\#} : \mathsf{PSh}(\mathsf{Set}) \to \mathsf{Sh}(\mathsf{Set})$ *that is left-adjoint to the inclussion Functor* $\iota : \mathsf{Sh}(\mathsf{Set}) \to \mathsf{PSh}(\mathsf{Set})$.

*Proof.* Let $\mathcal{F}$ be a Presheaf of sets over $X$. For every open subset $U \subseteq X$ we define

$$\mathcal{F}^{\#}(U) = \left\{ f_p \in \prod_{p \in U} \mathcal{F}_p \text{ verifying } (*) \right\}$$

where $(*)$ is the property *"for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of $p$ and a section $s \in \mathcal{F}(V)$ such that, for every $q \in V$, $s_q = f_q$"*.

Taking $\prod_{p \in U} \mathcal{F}_p \to \prod_{q \in V} \mathcal{F}_q$ as the restriction maps we see that $\mathcal{F}^{\#}$ is a Presheaf. Let's see now that $\mathcal{F}^{\#}$ is a Sheaf. To do this, consider an open subset $U \subseteq X$ with open cover $\{U_i\}_{i \in \mathcal{I}}$.

SH.1 Let $s, t \in \mathcal{F}^{\#}(U)$ be two sections such that, for all $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Then since for every $p \in U$ there exists $i \in \mathcal{I}$ such that $p \in U_i$ we have that $s_p = (s|_{U_i})_p = (t|_{u_i})_p = t_p$. Therefore $s = t$.

SH.2 Let $\{s_i \in \mathcal{F}^{\#}(U_i)\}_{i \in \mathcal{I}}$ be a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in \mathcal{I}$.

Consider $s \in \prod_{p \in U} \mathcal{F}_p$ defined as $s_p = (s_i)_p$ if $p \in U_i$. Note that this is well-defined by property $(*)$. To see that $s$ verifies $(*)$ note that for every $p \in U_i$ there exists an open neighbourhood $V \subseteq U$ of $p$ and $t \in \mathcal{F}(V)$ such that, for all $q \in V$, $t_q = (s_i)|_q = s_q$.

Let $V \subseteq U \subseteq X$ be two open sets. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \longrightarrow & \prod_{p \in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \longrightarrow & \prod_{q \in V} \mathcal{F}_q \end{array}$$

where the vertical arrows are the corresponding restriction maps. By chasing this diagram we get a map $\varphi : \mathcal{F} \to \mathcal{F}^{\#}$, $\varphi_U(s) = (s_p)_{p \in U}$.

It only remains to show that any morphism $\psi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a Sheaf of sets, factors uniquely as $\mathcal{F} \xrightarrow{\varphi} \mathcal{F}^{\#} \to \mathcal{G}$, i.e: we have to construct a morphism $\theta : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi = \theta \circ \varphi$.

Let $U \subseteq X$ be an open subset and $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ consider an open neighbourhood $V^{(p)} \subseteq U$ of $p$ and $t^{(p)} \in \mathcal{F}(V^{(p)})$ such that $t^{(p)}_q = s_q$. Consider now $\psi_{V^{(p)}}(t^{(p)}) \in \mathcal{G}(V^{(p)})$. Note that $\{\psi_{V^{(p)}}(t^{(p)})\}_{p \in U}$ is a compatible family of section in $\mathcal{G}$, therefore by the Sheaf axioms there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$. Therefore we can define $\theta_U(s) = t$.

By construction $\psi = \theta \circ \varphi$ and $\theta$ commutes with the restriction maps, making it a morphism of sheaves.

To check uniqueness, let $\theta' : \mathcal{F}^{\#} \to \mathcal{G}$ be another morphism such that $\psi = \theta' \circ \varphi$. Consider $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ choose $V^{(p)} \subseteq U$ and $s^{(p)} \in \mathcal{F}(V^{(p)})$ such that $s|_{V^{(p)}} = \varphi_{V^{(p)}}(s^{(p)})$. Then

$$\theta'(s)|_{V^{(p)}} = \theta'(s|_{V^{(p)}}) = \theta'(\varphi_{V^{(p)}}(s^{(p)})) = \psi_{V^{(p)}}(s^{(p)})$$

On the other hand $\theta(s)|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$ by definition. Therefore since $\mathscr{G}$ is a Sheaf by SH.1 we have $\theta'(f) = \theta(f)$, i.e: $\theta' = \theta$. □

We can extend the construction outlined in Theorem 1.2.1 to Algebraic Categorys.

**Lemma 1.2.1.** *Let $(\mathsf{C}, F)$ be an Algebraic Category. Let $\mathcal{F}$ be a $\mathsf{C}$-valued Presheaf. Then there exists a $\mathsf{C}$-valued Sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves $\mathcal{F} \to \mathcal{F}^{\#}$ satisfying*

1. *The map $\mathcal{F} \to \mathcal{F}^{\#}$ identifies the underlying Sheaf of sets of $\mathcal{F}^{\#}$ with the sheafification of the underlying Presheaf of sets of $\mathcal{F}$.*
2. *For any morphism of presheaves $\mathcal{F} \to \mathscr{G}$ where $\mathscr{G}$ is a $\mathsf{C}$-valued Sheaf there exists a unique factorization $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

*Remark* 1.2.1. We will assume that $\mathsf{C}$ is an Algebraic Category in what follows. ▽

Therefore we can assign a Sheaf to any Presheaf using this construction.

**Definition 1.2.4** (Sheafification)**.** Let $\mathcal{F}$ be a $\mathsf{C}$-valued Presheaf. We call the Sheaf $\mathcal{F}^{\#}$ obtained by Theorem 1.2.1 the **sheafification** of $\mathcal{F}$. ⌟

Using this adjunction we can *adapt* Definition 1.1.2 and Definition 1.1.3 to the case of Sheaves

**Definition 1.2.5** (Direct image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{F}$ be a $\mathsf{C}$-valued Sheaf over $X$. The **direct image Sheaf** of $\mathscr{F}$ via $f$ is the Sheaf $f_*\mathscr{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(U))$$

This construction defines a Functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

This means that Example 1.1.1 and Example 1.1.2 translate directly to Sheaves.

**Definition 1.2.6** (Global section Functor)**.** We denote by $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ the **global section functor** described in Example 1.1.1, but now with initial category $\mathsf{Sh}_X(\mathsf{C})$. ⌟

**Definition 1.2.7** (Skyscrapper Sheaf)**.** Let $X$ be a topological space and $p \in X$. Let $S$ be an object of $\mathsf{C}$. The **skyscrapper Sheaf** supported at $p$ $\iota_{*,p}\underline{S}$ is the Direct image Sheaf of the Constant Sheaf $\underline{S}$ via the map $\{p\} \to X$. ⌟

Rewrite this definitions

Explain why we need to sheafificate

**Definition 1.2.8** (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{G}$ be a $\mathsf{C}$-valued Sheaf over $Y$. The **inverse image Sheaf** of $\mathscr{G}$ via $f$ is the Sheaf $f^{-1}\mathscr{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathscr{G} = (f^{-1}\mathscr{G})^{\#}$$

where the RHS of the equation is the Sheafification of the Inverse image Presheaf of $\mathscr{G}$ via $f$.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

**Definition 1.2.9** (Constant Sheaf)**.** Let $S$ be an object of $\mathsf{C}$. The **constant Sheaf** $\underline{S}$ associated to $S$ is the Sheafification of the Constant Presheaf $\underline{S}_{\text{pre}}$, i.e: $\underline{S} = (\underline{S}_{\text{pre}})^{\#}$. ⌟

**Definition 1.2.10** (Stalk)**.** Let $X$ be a topological space and $p \in X$. Let $\mathscr{F}$ be a $\mathsf{C}$-valued Sheaf over $X$. The **stalk** of $\mathscr{F}$ at $p$ $\mathscr{F}_p$ is the Inverse image Sheaf of $\mathscr{F}$ via $X \to \{p\}$:

$$\mathscr{F}_p = \operatorname*{colim}_{U \ni p} \mathscr{F}(U)$$

⌟

Using the adjointness we have the analogous of Proposition 1.1.2

**Proposition 1.2.1.** *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Sheaf functor and the Inverse image Sheaf $f^{-1} \dashv f_*$.*

*Proof.*

$$\begin{aligned}\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) \\ &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G}) \\ &= \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G})\end{aligned}$$

□

Therefore adjunctions Corollary 1.1.1 and Corollary 1.2.2 also happen when considering Sheaves.

**Corollary 1.2.1.** *Let $X$ be a topological space and $p \in X$. Let $\mathscr{F}$ be a $\mathsf{C}$-valued Sheaves over $X$ and let $A$ be an object in $\mathsf{C}$. Then*

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\underline{A}, \mathscr{F}) \simeq \operatorname{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathscr{F}))$$

**Corollary 1.2.2.** *Let $X$ be a topological space and $p \in X$. Let $\mathscr{F}$ be a $\mathsf{C}$-valued Sheaves over $X$ and let $A$ be an object in $\mathsf{C}$. Then*

$$\operatorname{Hom}_{\mathsf{C}}(\mathscr{F}_p, A) \simeq \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F}, \iota_{p,*}\underline{A})$$

For cohomology we will be interested in the case $\mathsf{C} = \mathsf{Ab}$.

**Lemma 1.2.2.** *Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two Functors. Suppose that*

1. *$\mathsf{A}$ and $\mathsf{B}$ are additive categories, $a, b$ are additive functors and $b \dashv a$.*
2. *$\mathsf{B}$ is an Abelian category and $b$ is left-exact.*
3. *$ba \cong \mathrm{id}_{\mathsf{A}}$.*

*Then $\mathsf{A}$ is abelian.*

*Proof.* Since $\mathsf{B}$ is an Abelian category all Limits and Colimits exist en $\mathsf{B}$.

Since $b$ is left-adjoint it is right-exact and therefore, exact.

Let $\varphi : B_1 \to B_2$ be a morphism in $\mathsf{B}$. If $K = \ker(\varphi)$ then $K$ is the Equalizer of the diagram

$$K \longrightarrow B_1 \underset{0}{\overset{\varphi}{\rightrightarrows}} B_2$$

therefore applying $b$ we get

$$bK \longrightarrow bB_1 \underset{0}{\overset{b\varphi}{\rightrightarrows}} bB_2$$

i.e: $bK$ is the Kernel of $b\varphi$.

Dually we get that $b\operatorname{coker}(B_1 \to B_2)$ is the Cokernel of $b\varphi$.

Therefore we have that every morphism $b\varphi$ in A admits a Kernel and a Cokernel. Moreover since $ba \cong \mathrm{id}_{\mathsf{A}}$ every morphism in A is of the form $b\varphi$, therefore all Kernels and Cokernels exist in A. Indeed for every $\psi : A_1 \to A_2$ we have that

$$\ker \psi = b \ker(a\psi) \qquad \operatorname{coker} \psi = b \operatorname{coker}(a\psi) \tag{1.2.2}$$

It only remains to show that $\operatorname{coim} \psi = \operatorname{im} \psi$. Since A has all Limits and Colimits A has all finite Limits and Colimits, making $a$ left-exact. This means that $a$ transforms Kernels into Kernels. Therefore

$$\begin{aligned}
\operatorname{coim} \psi &= \operatorname{coker}(\ker \psi \to A_1) && \text{(definition)} \\
&= b \operatorname{coker}(a(\ker \psi \to A_1)) && \text{(1.2.2)} \\
&= b \operatorname{coker}(\ker(a\psi) \to aA_1) && (a \text{ preserves kernels}) \\
&= b \operatorname{coim}(a\psi) && \text{(definition)} \\
&= b \operatorname{im}(a\psi) && (\mathsf{B} \text{ is abelian}) \\
&= b \ker(aA_2 \to \operatorname{coker}(a\psi)) && \text{(definition)} \\
&= \ker(baA_2 \to b \operatorname{coker}(a\psi)) && (b \text{ preserves kernels}) \\
&= \ker(A_2 \to b \operatorname{coker}(a\psi)) && ba \cong \mathrm{id}_{\mathsf{A}} \\
&= \ker(A_2 \to \operatorname{coker} \psi) && \text{(1.2.2)} \\
&= \operatorname{im} \psi && \text{(definition)}
\end{aligned}$$

□

**Proposition 1.2.2.** *Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{Ab})$.*

1. *The Category $\mathsf{Sh}_X(\mathsf{Ab})$ is an Abelian category.*
2. *The Kernel of $\varphi$ is the same as the Kernel of $\varphi$ in $\mathsf{PSh}(\mathsf{Ab})$.*
3. *$\varphi$ is injective if and only if $\varphi$ is injective as a map or presheaves if and only if $\varphi$ is injective as a map of sheaves.*
4. *The Cokernel of $\varphi$ is the Sheafification of the Cokernel of $\varphi$ in $\mathsf{PSh}(\mathsf{Ab})$.*
5. *$\varphi$ is surjective if and only if $\varphi$ is surjective as a map of sheaves.*
6. *A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets $U$ and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$.*

*Proof.* Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 3., but this comes directly from Lemma 1.2.2 with $\mathbf{A} = \mathsf{Sh}_X(\mathsf{Ab})$, $\mathsf{B} = \mathsf{PSh}_X(\mathsf{Ab})$, $a : \mathsf{Sh}_X(\mathsf{Ab}) \hookrightarrow \mathsf{PSh}_X(\mathsf{Ab})$ the inclussion Functor and $b : \mathsf{PSh}_X(\mathsf{Ab}) \to \mathsf{Sh}_X(\mathsf{Ab})$ the Sheafification Functor. □

# Chapter 2

# Ringed spaces and $\mathscr{O}_X$-modules.

## 2.1 Ringed spaces

**Definition 2.1.1** (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where $X$ is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over $X$.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* $U$ and functions on $X$ are called (global) functions. ⌟

**Example 2.1.1.** Let and $X$ be a topological space. Then the $\mathbb{R}$-Algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$ with the usual restriction maps forms a Sheaf of rings over $X$. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

**Example 2.1.2.** If for every point $p \in X$ the Stalks $\mathscr{O}_{X,p}$ of the structure sheaf are Local rings then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

**Definition 2.1.2** (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is a `def:sheaf-morphism` of rings (i.e: a morphism in $\mathrm{Sh}_{\mathrm{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the Direct image Sheaf. ⌟

Maybe it is better to define $f^\sharp$ as an $f$-map. **[stacks-project]**

**Example 2.1.3.** Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by $f$ can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

## 2.2 $\mathscr{O}_X$ modules

**Definition 2.2.1** ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-Module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

**Example 2.2.1.** ... ◊

**Definition 2.2.2** (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a `def:sheaf-morphism` that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

**Example 2.2.2.** ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a Category $\mathscr{O}_X$-Mod.

### 2.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheaves of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules, where $\underline{\mathbb{Z}}_X$ is the `ex:constant-sheaf` with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over $X$ and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since $f$ is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of $U$ such that for every $i \in \mathcal{I} \quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every $U_i$ consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself $n_i$ times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of $x$ were $f$ is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that $f$ is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

## 2.3 $\mathscr{O}_X$-Mod is abelian.

**Definition 2.3.1** (Preadditive category)**.** A (small) category C is **preadditive** if every hom-set $\mathrm{Hom}(\triangle, \square)$ admits the structure of an abelian group and composition of morphisms is bilinear with respect to the group operation. ⌟

**Definition 2.3.2** (Additive category)**.** A category C is **additive** if it is a Preadditive category, has a zero object and has all binary biproducts. ⌟

**Definition 2.3.3** (Abelian category)**.** A Additive category C is **abelian** if

- It has all kernels and cokernels.
- Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

⌟

Move this definitions to the Homological Algebra section

Let's check first that $\mathscr{O}_X$-Mod is a Preadditive category.

**Lemma 2.3.1.** *The category $\mathscr{O}_X$-Mod is a Preadditive category.*

*Proof.* Let $\phi : \mathscr{F} \to \mathscr{G}$ and $\psi : \mathscr{F} \to \mathscr{G}$ be two Morphisms of Ringed spaces.

For each open subset $U \subseteq X$ define $(\phi + \psi)(U) = \phi(U) + \psi(U)$. □

Now Additive category

**Lemma 2.3.2.** *The category $\mathscr{O}_X$-Mod admits a zero object.*

*Proof.* Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

**Lemma 2.3.3.** *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

*Proof.* Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

**Theorem 2.3.1.** *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of $\varphi$ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker \varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

*Furthermore for every $x \in X$ $\quad (\ker \varphi)_x = \ker \varphi_x$.*

*Proof.* If $V \subseteq U$

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

□

**Theorem 2.3.2.** *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of $\varphi$ is the Sheafification of the quotient Presheaf*

$$(\mathrm{coker}_{\mathrm{pre}}\, \varphi)(U) = {}^{\mathscr{G}(U)}\big/_{\varphi(\mathscr{F}(U))}$$

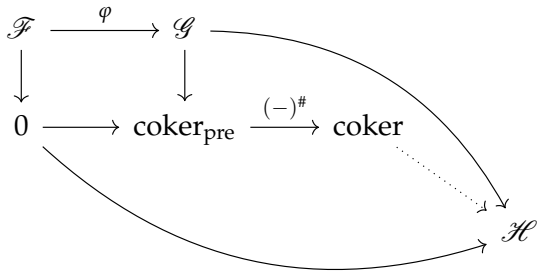
Prove that this is the cokernel in PSh.

*Furthermore for every $x \in X$ $\quad (\mathrm{coker}\, \varphi)_x = \mathrm{coker}\, \varphi_x$.*

*Proof.* Given a commutative diagram

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\varphi} & \mathscr{G} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathscr{H} \end{array}$$

We construct




□

Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian category.

## 2.4 Sequences of $\mathscr{O}_X$-Mod

**Theorem 2.4.1.** *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

# Appendix A

# Category Theory

**Definition A.0.1** (Category)**.** A **category** $\mathsf{C}$ consists of:

- A collection of *objects* $X, Y, \ldots$
- For each pair of objects $X, Y$ a set of morphisms $\mathrm{Hom}_{\mathsf{C}}(X, Y)$. We denote this morphisms by $f : X \to Y$.

such that

- Each object has an *identity morphism* $1_X : X \to X$.
- For each pair of morphisms $f : X \to Y, g : Y \to Z$ there exists a *composite morphism* $gf : X \to Z$.

We impose two axioms on this data

1. For any $f : X \to Y$ the composites $1_Y f$ and $f 1_X$ are equal to $f$.
2. For any triplet of morphisms $f : X \to Y, g : Y \to W, h : W \to Z$ the composites $h(gf)$ and $(hg)f$ are equal. We denote this composite by $hgf$.

⌟

*Remark* A.0.1. Technically speaking Definition A.0.1 is the definition of a *locally small category*. In general, a category $\mathsf{C}$ can have an arbitrarily *large* collection of morphisms between two objects. ▽

Every Category has an associated opposite category

**Definition A.0.2** (Opposite Category)**.** Let $\mathsf{C}$ be a Category. The **opposite Category** of $\mathsf{C}$, $\mathsf{C}^{\mathrm{op}}$ has

- The same objects as $\mathsf{C}$.
- For every pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

⌟

**Definition A.0.3.** A morphism $f : x \to y$ in a Category $\mathsf{C}$ is **monomorphism** if for any parallel pair of morphisms $h, k : w \rightrightarrows x, fh = fk$ implies $h = k$. Dually, $f$ is an **epimorphism** if for any pair of morphisms $h, k : y \rightrightarrows z, hf = kf$ implies $h = k$. ⌟

*Notation.* We denote monomorphisms by $f : x \rightarrowtail y$ and epimorphisms by $f : x \twoheadrightarrow y$.

**Definition A.0.4** (Isomorphism)**.** A morphism $f : x \to y$ in a Category $\mathsf{C}$ is an **isomorphism** if there exists a morphism $g : y \to x$ in $\mathsf{C}$ such that $fg = 1_y$ and $gf = 1_x$.

The objects $x$ and $y$ are called *isomorphic* if there exists an isomorphism between them, and we denote this by $x \cong y$. ⌟

**Definition A.0.5** (Functor)**.** A **functor** $F : \mathsf{C} \to \mathsf{D}$ between two Categories consists of:

- For each object $c$ in $\mathsf{C}$ an object $Fc \in \mathsf{D}$.
- For each morphism $f : c \to c'$ in $\mathsf{C}$ a morphism $Ff : Fc \to Fc'$ in $\mathsf{D}$.

satisfying the following axioms

1. For each element $c$ in $\mathsf{C}$, $F1_c = 1_{Fc}$.
2. For any composable pair of morphisms $f, g$ in $\mathsf{C}$, $Fg \cdot Ff = F(g \cdot f)$.

⌟

**Example A.0.1** (Posetal category of open sets)**.** The Functor $\mathsf{Op} : \mathsf{Top} \to \mathsf{Poset}$ sends a topological space $X$ to its poset of open sets.

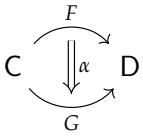
Therefore for every topological space with have an associated *posetal category of open sets* $\mathsf{Op}(X)$. ♢

**Definition A.0.6** (Natural transformation)**.** Let $\mathsf{C}, \mathsf{D}$ be two Categories and $F, G : \mathsf{C} \rightrightarrows \mathsf{D}$ be two Functors. A **natural transformation** $\alpha : F \Rightarrow G$ consists of:

- For each object $c \in \mathsf{C}$ an arrow $\alpha_c : Fc \to Gc$ in $\mathsf{D}$ such that, for any morphism $f : c \to c'$ in $\mathsf{C}$ the following diagram commutes in $\mathsf{D}$

$$\begin{array}{ccc} Fc & \xrightarrow{\alpha_c} & Gc \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ Fc' & \xrightarrow[\alpha_{c'}]{} & Gc' \end{array}$$

We denote this by




⌟

**Definition A.0.7.** A Natural transformation $\alpha : F \Rightarrow G$ is a **natural isomorphism** if every component $\alpha_f$ is an Isomorphism.

In this case, we say that $F$ and $G$ are **naturally isomorphic** and we denote it by $\alpha : F \cong G$. ⌟

**Definition A.0.8.** A Functor $F : \mathsf{C} \to \mathsf{D}$ is

- **full** if for all pair of objects $x, y \in \mathsf{C}$ the map $\mathrm{Hom}_{\mathsf{C}}(x, y) \to \mathrm{Hom}_{\mathsf{D}}(Fx, Fy)$ is surjective.
- **faithful** if for all pair of objects $x, y \in \mathsf{C}$ the map $\mathrm{Hom}_{\mathsf{C}}(x, y) \to \mathrm{Hom}_{\mathsf{D}}(Fx, Fy)$ is injective.
- **essentially surjective on objects** if for every object $d \in \mathsf{D}$ there exists $c \in \mathsf{C}$ such that $d \cong Fc$.

If $F$ is both full and faithfull we will say it is **fully faithful**. ⌟

**Definition A.0.9** (Concrete category)**.** A Category $\mathsf{C}$ is a **concrete Category** if there exists a Functor $F : \mathsf{C} \to \mathsf{Set}$ that is faithful. ⌟

**Definition A.0.10** (Diagram)**.** A **diagram** in a Category $\mathsf{C}$ with shape $\mathsf{J}$ is a Functor $F : \mathsf{C} \to \mathsf{J}$. ⌟

**Definition A.0.11** (Initial and terminal objects)**.** Let $\mathsf{C}$ be a Category. An object $i \in \mathsf{C}$ is **initial** if for every object $c \in \mathsf{C}$ there exists a unique morphism $i \to c$. Dually, an object $t \in \mathsf{C}$ is **terminal** if for every object $c \in \mathsf{C}$ there exists a unique morphism $c \to t$. ⌟

**Definition A.0.12.** Let $c$ be an object in a Category $\mathsf{C}$. The **Functor represented** by $c$ is the Functor $\mathrm{Hom}_{\mathsf{C}}(c, -) \to \mathsf{Set}$ defined as

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}}(c,x) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f_*} \\ y & & \mathrm{Hom}_{\mathsf{C}}(c,y) \end{array}$$

⌟

*Remark* A.0.2. If we consider the Opposite Category $\mathsf{C}^{\mathrm{op}}$ in Definition A.0.12 for each $c \in \mathsf{C}^{\mathrm{op}}$ we get a Functor $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c, -) \to \mathsf{Set}$ defined as

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,x) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f_*} \\ y & & \mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c,y) \end{array}$$

Therefore since $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(c, x) = \mathrm{Hom}_{\mathsf{C}}(x, c)$ we get a functor $\mathrm{Hom}_{\mathsf{C}}(-, c) \to \mathsf{Set}$

$$\begin{array}{ccc} x & & \mathrm{Hom}_{\mathsf{C}}(x,c) \\ {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle f^*} \\ y & & \mathrm{Hom}_{\mathsf{C}}(y,c) \end{array}$$

▽

**Definition A.0.13.** A Functor $F : \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists $c \in \mathsf{C}$ such that $F$ is naturally isomorphic to the represented functor associated to $c$. ⌟

**Definition A.0.14** (Limit)**.** ⌟

**Definition A.0.15** (Colimit)**.** ⌟

**Definition A.0.16** (Equalizer)**.** ⌟

**Definition A.0.17** (Adjoint)**.** ⌟

**Definition A.0.18** (Product)**.** ⌟

**Definition A.0.19** (Coproduct)**.** ⌟

**Definition A.0.20.** ⌟

**Definition A.0.21** (Subcategory)**.** ⌟

**Definition A.0.22** (Full Subcategory)**.** ⌟

**Definition A.0.23** (Directed Colimit)**.** ⌟

**Proposition A.0.1** (Functors preserve Isomorphisms)**.**

**Definition A.0.24** (Universal property)**.** ⌟

**Lemma A.0.1** (Yonneda lemma)**.**

**Definition A.0.25** (Algebraic Category)**.** ⌟

Move

**Definition A.0.26** (Abelian category)**.** ⌟

**Definition A.0.27** (Kernel)**.** ⌟

**Definition A.0.28** (Cokernel)**.** ⌟

# Appendix B

# Algebra

**Definition B.0.1** (Algebra)**.** ⌟

**Definition B.0.2** (Local ring)**.** ⌟

**Definition B.0.3** (Module)**.** ⌟

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